

Calc 1 Answer sheet

August 22, 2025

Practice

$$\frac{d}{dx}4x^3 = 12x^2$$

$$\frac{d}{dx}(x^2 + 2x) = 2(x + 1)$$

More practice

$$1) \frac{d}{dx}2x^2 + 10 = 4x$$

$$2) \frac{d}{dx}5x^4 - \frac{2}{3}x^3 = 2x^2(10x - 1)$$

$$3) \frac{d}{dx}9\sqrt{x} = \frac{9}{2\sqrt{x}}$$

$$4) \frac{d}{dx}\frac{4}{x^2} = -\frac{8}{x^3}$$

$$5) \frac{d}{dx}\frac{2w}{5} = 0$$

More more practice

$$1) \frac{d}{dx}(10e^x) = 10e^x$$

$$2) \frac{d}{dx}(\ln(x) - \frac{e^2}{3}) = \frac{1}{x}$$

More more more practice

$$1) \frac{d}{dx}x^3 \cdot x = 4x^3$$

$$2) \frac{d}{dx}e^x \cdot x^2 = e^x x(x + 2)$$

$$3) \frac{d}{dx}(3x^4 - 8)^2 = 24x^3(3x^4 - 8)$$